

Full length article

Experimental and numerical investigations of hot-rolled stainless steel channel section columns susceptible to flexural buckling

Shuai Li^a, Lulu Zhang^a, Yating Liang^b, Ou Zhao^{a,*}

^a School of Civil and Environmental Engineering, Nanyang Technological University, Singapore

^b School of Engineering, University of Glasgow, Glasgow, UK



ARTICLE INFO

Keywords:

Design analyses
Flexural buckling
Hot-rolled stainless steel channel sections
Membrane residual stress measurements
Numerical modelling
Pin-ended column tests

ABSTRACT

A thorough testing and numerical modelling programme has been conducted to investigate the minor-axis flexural buckling behaviour and resistances of pin-ended hot-rolled stainless steel channel section columns. The testing programme included material coupon tests, membrane residual stress measurements, initial global and local geometric imperfection measurements and ten pin-ended channel section column tests about the minor principal axis. A membrane residual stress predictive model was developed based on the measured data. The testing programme was followed by a numerical modelling programme; finite element models were developed to simulate the test results and then adopted to perform parametric studies to generate further numerical data on hot-rolled stainless steel channel section columns over a wide range of cross-section dimensions and member effective lengths. The obtained test and numerical data were then adopted to assess the accuracy of relevant design rules for hot-rolled stainless steel channel section columns failing by flexural buckling about the minor principal axis, as given in the European code and American design guide. The assessment results revealed that (i) the European code yields many unsafe flexural buckling resistance predictions and (ii) the American design guide yields overly conservative and scattered flexural buckling resistance predictions. Finally, a revised Eurocode buckling curve was proposed and shown to offer accurate, consistent and safe-sided flexural buckling resistance predictions for hot-rolled stainless steel channel section columns over the whole considered range of member non-dimensional slendernesses. The reliability of the revised Eurocode buckling curve was also confirmed using statistical analyses based on EN 1990.

1. Introduction

Stainless steel offers numerous advantages over carbon steel, including superior corrosion resistance, high material strength and exceptional ductility [1,2]. Therefore, the use of stainless steel members in civil and offshore engineering becomes increasingly prevalent. Channel sections with simple geometry are widely used in structural applications. For example, channel sections are used as rafters spanning from eaves to ridges in lightweight frames, as bracing members in low-rise and medium-rise buildings to resist lateral forces and as frameworks to position wooden boards in wood-steel composite structures.

Previous experimental and numerical studies on stainless steel channel section members are briefly reviewed herein. At cross-sectional level, Kuwamura [3], Dobrić et al. [4] and Zhang et al. [5] studied the compression resistances of press-braked stainless steel channel section stub columns based on tests and numerical modelling, while the in-plane flexural resistances of cold-formed and laser-welded stainless steel channel section beams were experimentally and numerically investigated by Zhang et al. [5] and Theofanous et al. [6], respectively.

Liang et al. [7–9] conducted systematic testing and numerical modelling programmes to investigate the cross-sectional instability and resistances of laser-welded stainless steel channel section stub columns under combined compression and bending. All the aforementioned studies highlighted that the current design codes result in unduly conservative cross-section resistance predictions for stainless steel channel sections under isolated and combined loadings, owing to the use of 0.2% proof stress as the design stress without considering material strain hardening. At member level, pin-ended and fixed-ended compression tests on cold-formed stainless steel lipped channel section columns were conducted by Becque and Rasmussen [10] and Rossi et al. [11], to investigate their local–overall interactive buckling behaviour and flexural-torsional–distortional interactive buckling behaviour, respectively. Dobrić et al. [12] and Zhang et al. [13] experimentally and numerically investigated the flexural buckling behaviour of cold-formed stainless steel plain channel section columns, pointed out the inaccuracy and shortcomings of the codified design methods and proposed new improved design approaches.

* Corresponding author.

E-mail address: ou.zhao@ntu.edu.sg (O. Zhao).

<https://doi.org/10.1016/j.tws.2021.107791>

Received 31 October 2020; Received in revised form 13 March 2021; Accepted 1 April 2021

Available online 23 April 2021

0263-8231/© 2021 Elsevier Ltd. All rights reserved.

It is worth highlighting that all of the aforementioned studies are on stainless steel channel section members fabricated by welding and cold-forming. Recently, with the advancement in manufacturing and metallurgical techniques, direct hot-rolling of stainless steel channel sections becomes possible [14,15]. In comparison with welding and cold-forming, hot-rolling leads to lower fabrication costs and therefore has a rather good potential in fabricating stainless steel cross-sections, especially in view of the fact that the material cost of stainless steel is high. From the viewpoint of mechanical characteristics, hot-rolled stainless steel channel sections have a much lower level of membrane residual stresses than their welded counterparts and more homogeneous material properties and more consistent hardness than their cold-formed counterparts.

As part of an ongoing research project on hot-rolled stainless steel channel section members, this study is focusing on the flexural buckling behaviour and resistances of hot-rolled stainless steel channel section columns. In the present paper, a testing programme, including material coupon tests, membrane residual stress measurements, initial global and local geometric imperfection measurements and pin-ended column tests on ten hot-rolled stainless steel channel section column specimens, was firstly conducted. Following the testing programme, a numerical modelling programme was conducted, including a validation study, where finite element models were developed and validated against the test results, and a parametric study, where the validated finite element modes were used to generate further numerical data over a wide range of cross-section dimensions and member effective lengths. The obtained test and numerical data were used to assess the accuracy of the codified design rules, as given in EN 1993-1-4 [16] and AISC 27 [17], for hot-rolled stainless steel channel section columns prone to flexural buckling about the minor principal axis, followed by the development of a new design method.

2. Testing

2.1. Test specimens

Two plain channel sections C 80 × 40 × 5 and C 100 × 50 × 5, hot-rolled from grade EN 1.4301 austenitic stainless steel plates, were adopted in the testing programme, and it is worth noting that C 80 × 40 × 5 and C 100 × 50 × 5 are respectively categorised as Class 2 and Class 3 based on the slenderness limits given in EN 1993-1-4 [16] and also fall in the category of non-slender cross-sections according to the limiting width-to-thickness ratios specified in AISC 27 [17]. For each adopted channel section, five column specimens were prepared with different member lengths, leading to a total of ten column specimens in the testing programme. The identifier of each specimen includes the cross-section label 'C1' or 'C2' ('C1' signifying channel section C 80 × 40 × 5 and 'C2' representing channel section C 100 × 50 × 5), a letter 'L' (indicating length) and a number (for the purpose of differentiating specimens with different member lengths but the same cross-section size), e.g., C2-L2. The measured geometric dimensions of each hot-rolled stainless steel channel section column specimen are reported in Table 1, where L is the column length, B_w is the outer web width, B_f is the outer flange width, t is the wall thickness and r is the corner inner radius - see Fig. 1, where the notations of cross-section geometric parameters for hot-rolled channel sections are shown.

2.2. Material tests

Prior to pin-ended column tests, tensile coupon tests were carried out to obtain the material properties of the two adopted hot-rolled stainless steel channel sections. For each channel section, three coupons were extracted from the middle parts of the web and flanges in the longitudinal direction - see Fig. 1, and the dimensions of the coupons were machined in accordance with the requirements given in the European code EN ISO 6892-1 [18], as illustrated in Fig. 2(a). An INSTRON

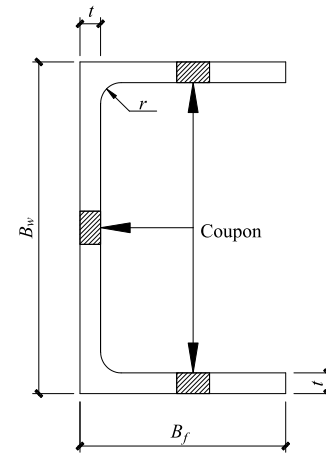


Fig. 1. Notations of geometric parameters of hot-rolled channel sections and locations of coupons.

250 kN hydraulic testing machine was adopted to conduct the material tensile coupon tests under displacement control, with the rates equal to 0.05 mm/min and 0.8 mm/min before and after the material 0.2% proof stress, respectively. The material tensile coupon test setup is shown in Fig. 2(b), where two strain gauges are attached to the mid-height of the coupon to measure tensile strains in the longitudinal direction and an extensometer is mounted onto the necked part of the coupon to record the elongation over the gauge length of 50 mm. The label of each coupon starts with the channel section identifier and ends with the location indicator within the channel section ('F1', 'F2' and 'W' signifying the flanges and web, respectively). The key obtained material properties are reported in Table 2, where E is the Young's modulus, $\sigma_{0.2}$ is the 0.2% proof stress, σ_u is the ultimate stress, ϵ_u is the strain at the ultimate stress, ϵ_f is the strain at fracture, and n and m are the parameters used in the Ramberg–Osgood material model [19–23]. Fig. 3(a)–(b) show the measured material stress–strain curves for the two adopted channel sections.

2.3. Initial global and local geometric imperfection measurements

Initial geometric imperfections naturally exist in metallic channel section members and are known to affect their buckling behaviour and resistances [13,24–28]. Therefore, the initial global and local geometric imperfections of each hot-rolled stainless steel channel section column specimen were carefully measured. The test setup for initial global geometric imperfection measurements is shown in Fig. 4, where a column specimen is mounted onto the flat work bench of a CNC router, and an LVDT, pointing to the web of the column specimen, is moved along the centreline to measure the deviations about the minor principal axis (i.e. the axis of flexural buckling). The initial mid-height global geometric imperfection amplitude ω_g of each specimen was then determined as the deviation from a linear reference line (i.e. a straight line connecting the data points recorded by the LVDT at the two ends) to the data point recorded at the mid-height, as reported in Table 1. It is worth noting that the obtained initial global geometric imperfection amplitude is taken as positive if the column specimen initially bows towards the flange tips - see Fig. 5(a), but negative if the column specimen initially bows towards the web - see Fig. 5(b). The test setup for initial local geometric imperfection measurements of a column specimen is similar to that adopted for initial global geometric imperfection measurements, as shown in Fig. 6. It is worth mentioning that the initial local geometric imperfection of each specimen was measured along the centrelines of the three constituent plates (one web and two flanges) over the central 300 mm; this length is regarded short enough to preclude the influence from initial global geometric

Table 1

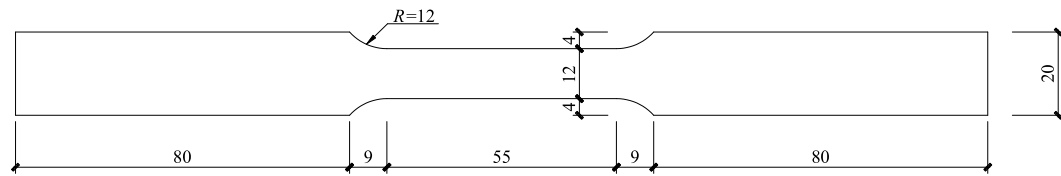
Measured geometric dimensions and initial geometric imperfections of hot-rolled stainless steel channel section column specimens.

Cross-section	Specimen ID	L (mm)	B_w (mm)	B_f (mm)	t (mm)	r (mm)	ω_0 (mm)	ω_g (mm)
C 80 × 40 × 5	C1-L1	397.4	80.47	40.28	4.73	4.70	0.03	0.13
	C1-L2	598.2	80.33	40.66	4.76	4.70	0.02	0.16
	C1-L3	799.2	80.41	40.28	4.82	4.74	0.03	−0.09
	C1-L4	998.3	80.41	40.07	4.73	4.69	0.03	−0.22
	C1-L5	1200.8	80.45	40.07	4.75	4.71	0.04	−0.32
C 100 × 50 × 5	C2-L1	500.0	98.99	49.61	4.91	4.85	0.03	0.06
	C2-L2	698.5	99.01	49.68	4.82	4.82	0.04	−0.1
	C2-L3	896.3	99.08	49.70	4.96	4.90	0.03	−0.12
	C2-L4	1120.8	99.12	49.91	4.74	4.71	0.03	0.50
	C2-L5	1299.5	99.11	49.82	4.86	4.82	0.03	0.24

Table 2

Summary of key measured material properties from tensile coupon tests.

Cross-section	Coupon ID	E	$\sigma_{0.2}$	σ_u	ϵ_u	ϵ_f	R-O parameters	
		(GPa)	(MPa)	(MPa)	(%)	(%)	n	m
C 80 × 40 × 5	C1-W	205	363	700	53.0	66.9	6.4	2.4
	C1-F1	200	354	757	58.0	75.0	8.1	1.6
	C1-F2	202	352	721	58.6	69.4	7.4	2.1
C 100 × 50 × 5	C2-W	204	352	707	50.0	62.1	8.4	2.5
	C2-F1	202	327	743	57.7	73.0	12.8	2.4
	C2-F2	203	316	741	60.0	65.4	11.7	1.9



(a) Coupon dimensions.



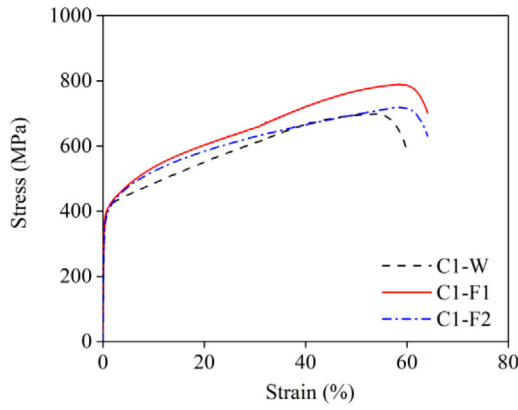
(b) Test setup.

Fig. 2. Material tests.

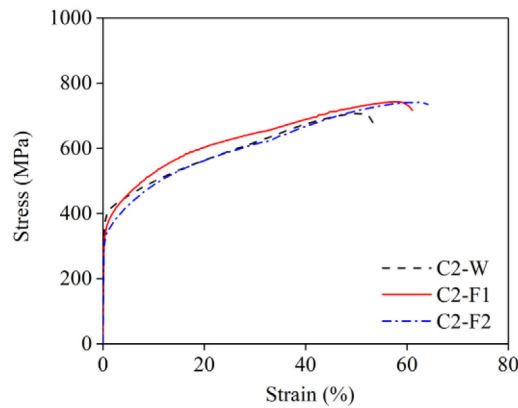
imperfection, but still long enough to incorporate a typical distribution pattern of initial local geometric imperfection. For each constituent plate of a channel section column specimen, the initial local geometric imperfection magnitude was taken as the maximum deviation of the measured data set from a linear regression line fitted to the measured data set, while the initial local geometric imperfection magnitude of a channel section column specimen ω_0 was defined as the maximum of the initial local geometric imperfection magnitudes measured from all the three constituent plates, as reported in Table 1.

2.4. Membrane residual stress measurements

Membrane residual stress measurements were conducted on two large-sized (representative) hot-rolled stainless steel channel sections C 100 × 50 × 5 and C 150 × 75 × 6, based on the sectioning method and the recommended procedures given in Ziemian [29]. Figs. 7 and 8 display the locations and dimensions of strips within the two representative hot-rolled stainless steel channel sections before sectioning; the nominal lengths of the strips are 150 mm, with the widths ranging from 6 mm to 10 mm (depending on regions). Prior to sectioning, two gauge holes with a diameter of 1.98 mm were drilled along the centreline



(a) C 80x40x5



(b) C 100x50x5

Fig. 3. Measured material stress-strain curves.

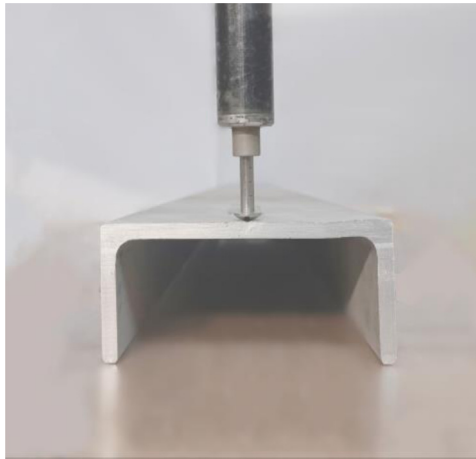


Fig. 4. Test setup for initial global geometric imperfection measurements.

of the outer face of each strip and at a distance of 25 mm from the strip ends, through the use of an automatic dot puncher, leading to the nominal length between the two gauge holes L_0 equal to 100 mm; the actual strip length between the two gauge holes was then measured by using a Demec gauge. Upon length measurements of the strips within channel sections, each specimen was cut into strips, allowing for the release of membrane residual stresses; a typical sectioned C 150 $\times$ 75 $\times$ 6 specimen is shown in Fig. 9. The Demec gauge was again adopted to

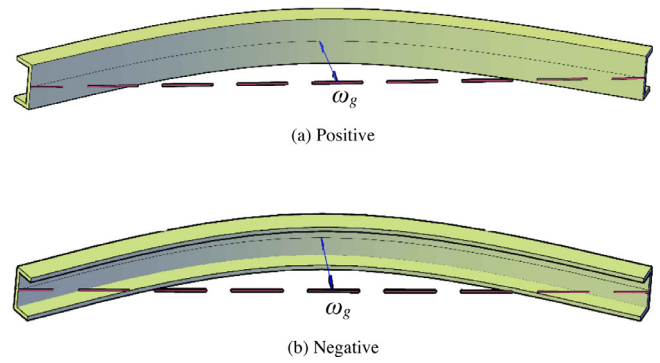
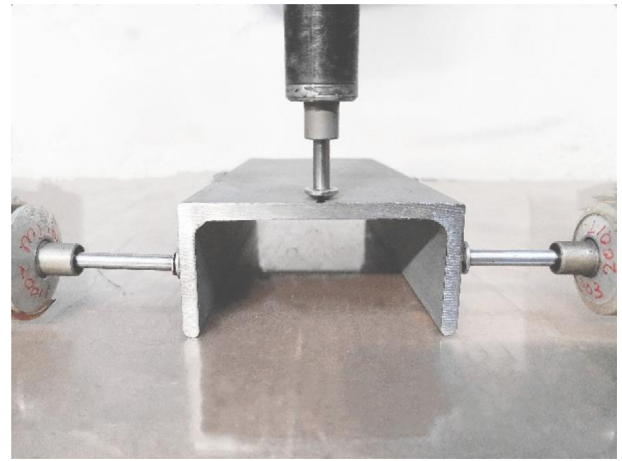
Fig. 5. Sign convention of initial global geometric imperfection ω_g .

Fig. 6. Test setup for initial local geometric imperfection measurements.

measure the strip lengths after sectioning. For each strip, the readings of the Demec gauge taken before and after sectioning are denoted as r_1 and r_2 , respectively. To eliminate the influence of temperature changes during the membrane residual stress measurements, a temperature reference bar, made of the same material as the hot-rolled stainless steel channel section specimens, was used, with the lengths measured before and after the sectioning process denoted as t_1 and t_2 , respectively.

On the basis of the readings obtained from the Demec gauge and temperature reference bar before and after sectioning, the relieved membrane residual strains can be calculated from Eq. (1); note that positive (calculated) strains signify tensile membrane residual stresses, while negative strains determined from Eq. (1) indicate compressive membrane residual stresses. For strips within the web-to-flange junction regions, where high through-thickness residual stress gradients exist and result in slightly curved strips after sectioning, the relieved membrane residual strains calculated from Eq. (1) were further corrected according to Eq. (2) [30] to eliminate the influence of longitudinal curvature, where δ is the maximum deviation from a straight reference line connecting the two gauge holes of the strip.

$$\epsilon = \frac{(r_1 - t_1) - (r_2 - t_2)}{L_0 + (r_1 - t_1)} \quad (1)$$

$$\epsilon_c = \epsilon + \frac{(\delta/L_0)^2}{6(\delta/L_0)^4 + 1} \quad (2)$$

Upon determination of the relieved membrane residual strains for all the strips, the corresponding membrane residual stresses were then calculated through multiplying the relieved membrane residual strains by the Young's moduli obtained from the material tensile coupon tests, as shown in Fig. 10(a) and (b) for the two hot-rolled stainless steel

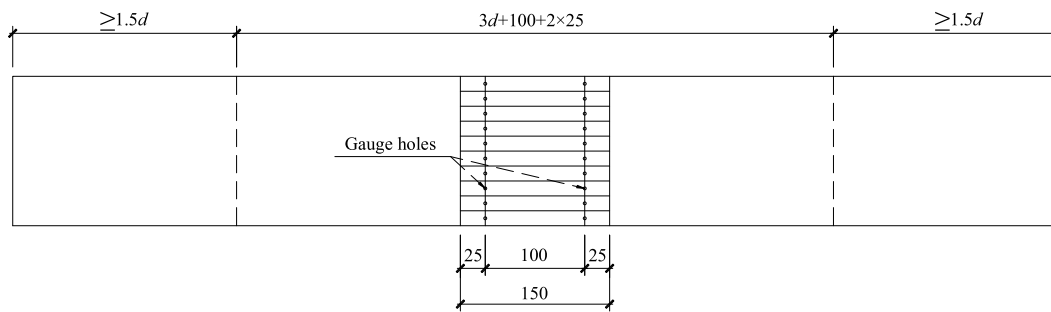


Fig. 7. Locations of strips cut for membrane residual stress measurements (in mm).

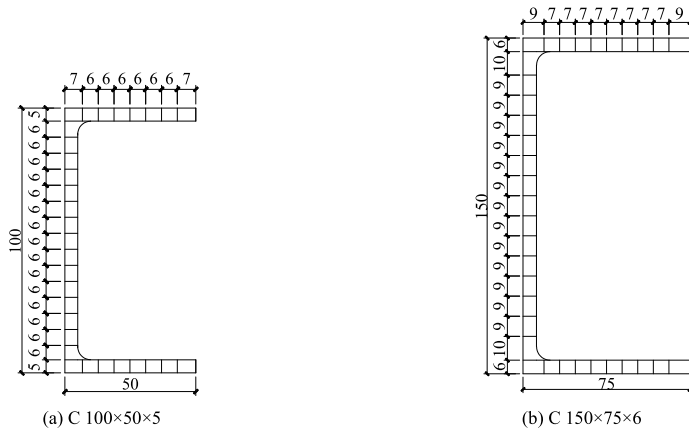


Fig. 8. Locations and dimensions of strips within channel sections (in mm).

Table 3
Measured peak membrane residual stresses in hot-rolled stainless steel channel sections.

Cross-section	$\sigma_{M1}/\sigma_{0.2}$	$\sigma_{F1}/\sigma_{0.2}$	$\sigma_{WC}/\sigma_{0.2}$	$\sigma_{FC}/\sigma_{0.2}$
C 100 × 50 × 5	0.20	0.18	-0.23	-0.16
	0.22	0.25	-0.23	-0.24
C 150 × 75 × 6	0.23	0.26	-0.20	-0.22
	0.25	0.17	-0.20	-0.16

channel sections C 100 $\times$ 50 $\times$ 5 and C 150 $\times$ 75 $\times$ 6; note that positive and negative values signify tensile and compressive membrane residual stresses, respectively. The obtained maximum tensile and compressive membrane residual stresses were normalised by the corresponding measured yield stresses (the 0.2% proof stresses), as reported in [Table 3](#), where the subscripts 'w' and 'f' represent the web and flange, respectively, while the subscripts 'c' and 't' denote the compressive and tensile membrane residual stresses, respectively. [Fig. 11\(a\)](#) and [\(b\)](#) show the measured membrane residual stress patterns and amplitudes in a normalised format; note that the normalised position at the origin point 0.0 corresponds to the web-to-flange junction, while the normalised position at point 1.0 represents the mid-point of the web or the flange tip. On the basis of the experimental data, a predictive model, as shown in [Fig. 12](#), was proposed for predicting membrane residual stresses in hot-rolled stainless steel channel sections, with linear distribution patterns for the flange and half of the web and the maximum compressive (or tensile) membrane residual stress magnitudes equal to $0.3\sigma_{0.2}$. The proposed predictive model was shown to agree well with the measured membrane residual stress data - see [Fig. 11](#).

2.5. Pin-ended column tests

Compression tests on ten pin-ended hot-rolled stainless steel channel section columns were conducted to study their minor-axis flexural

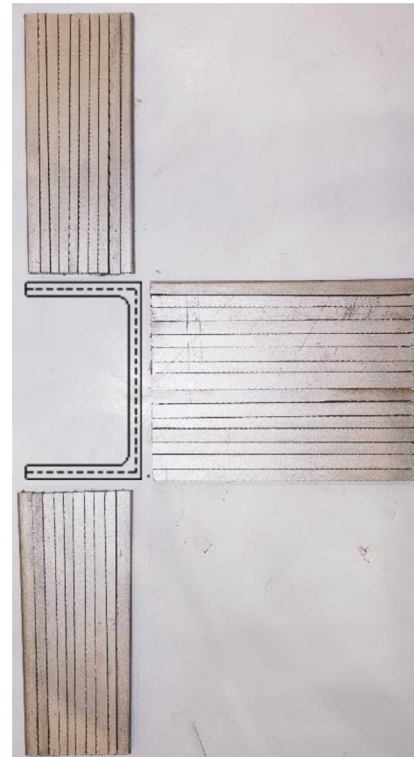


Fig. 9. Typical sectioned C 150 $\times$ 75 $\times$ 6 specimen.

buckling behaviour and resistances. The column specimen ends were welded with 15-mm-thick plates, which helped prevent end cross-sections from warping and twisting. An INSTRON 5000 kN servo-hydraulic testing machine was adopted to apply compression loads onto the column specimens and a displacement-control method was used to drive the testing machine, with the rate equal to 0.2 mm/min. A photograph and a schematic diagram of the pin-ended column test setup are shown in Fig. 13. Each end of the testing machine was equipped with a knife-edge device, including a pit plate with a semi-circular groove and a wedge plate with a knife-edge wedge, to offer pin-ended boundary conditions to the column specimen ends about the axis of buckling (i.e. the minor principal axis). Prior to testing, a column specimen was firstly positioned between the top and bottom wedge plates, with the cross-section minor principal axis parallel with the knife edges and the member longitudinal axis intersecting with the knife edges at right angles, and then anchored at both ends by stiffening plates (bolted to the wedge plates using high strength bolts), before placed between the top and bottom pit plates in the testing machine. Note that the distance from the rotation centre of the knife-edge device to the end section of the specimen is equal to 75 mm, resulting in the effective length of each column specimen L_e equal to the measured

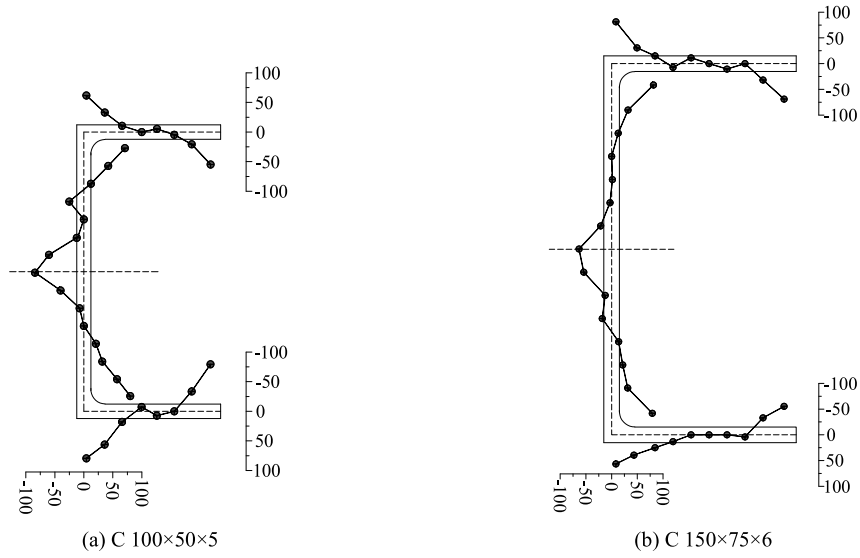
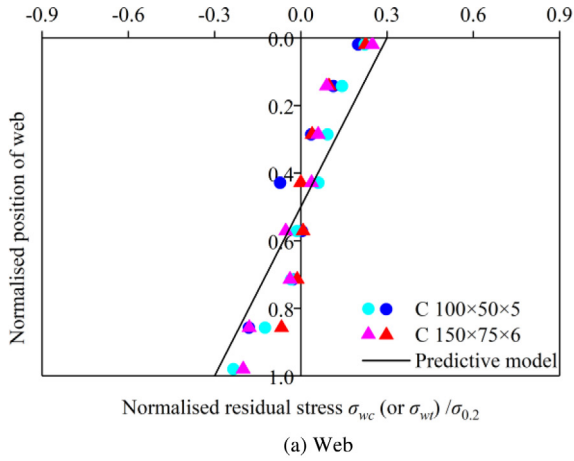
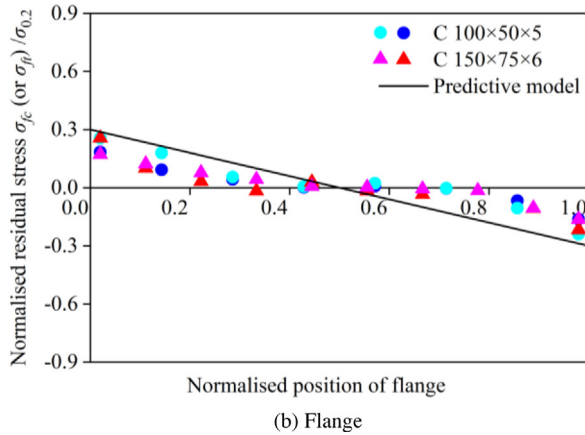


Fig. 10. Measured membrane residual stress patterns and amplitudes (in MPa).



(a) Web



(b) Flange

Fig. 11. Comparison between measured membrane residual stress patterns and amplitudes and predictive model.

member length plus an additional length of 150 mm, and the member non-dimensional slenderness about the minor principal axis (i.e. the

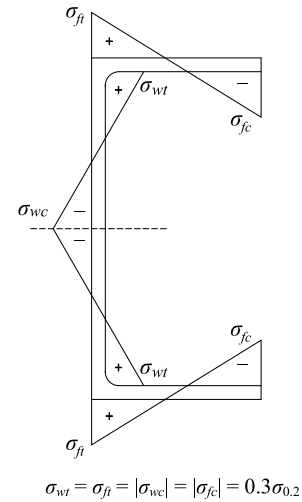


Fig. 12. Membrane residual stress predictive model for hot-rolled stainless steel channel sections.

axis of buckling) is calculated from Eq. (3), where A is the cross-section area and I is the second moment of area about the minor principal axis.

$$\bar{\lambda} = \sqrt{\frac{A\sigma_{0.2}L_e^2}{\pi^2 EI}} \quad (3)$$

With regard to the instrumentation employed for the pin-ended column tests, as shown in Fig. 13, an LVDT, horizontally pointing to the centreline of the web at mid-height, was adopted to monitor the lateral deflection of the column specimen along the buckling direction and two strain gauges, attached to the outer faces of the web and flange tip at the mid-height of the column specimen, were used to record the longitudinal strains at the two positions. For each column specimen, the readings from the LVDT and strain gauges were used to determine its overall global geometric imperfection amplitude e_m , defined as the summation of the initial loading eccentricity e_0 and the initial mid-height global geometric imperfection amplitude ω_g , based on Eq. (4) [13,31–33], where $\epsilon_{max} - \epsilon_{min}$ is the difference of the longitudinal strains measured by the one pair of strain gauges, N is the applied compression load, d_s is the distance between the two strain gauges

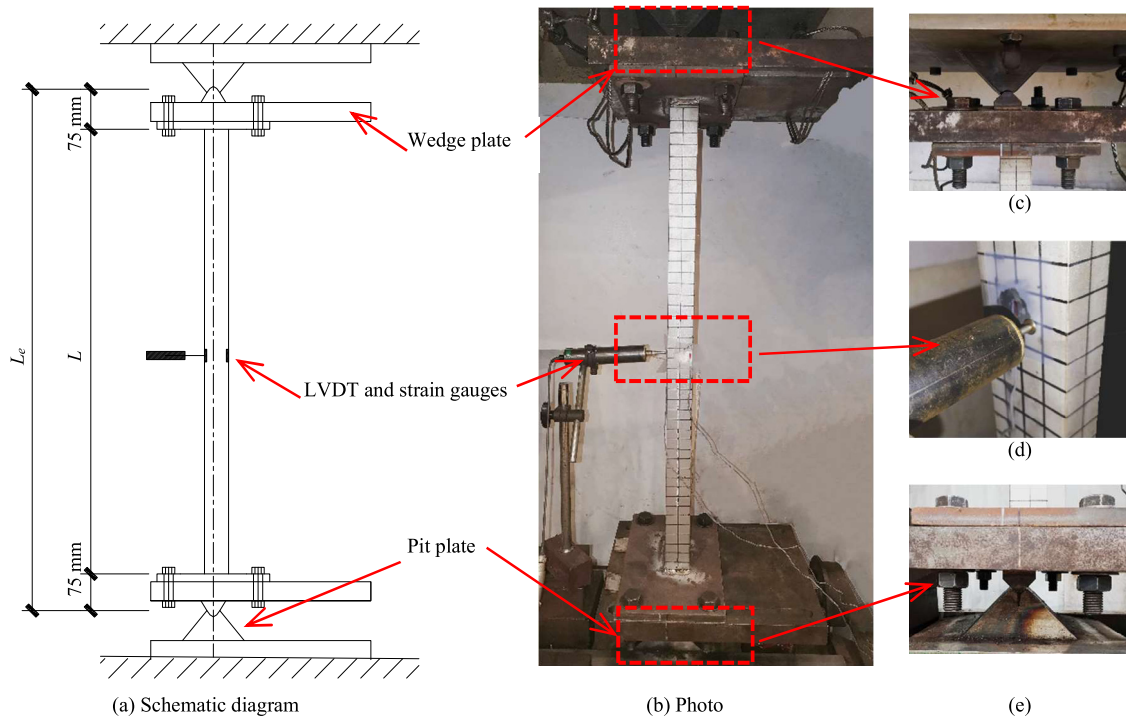
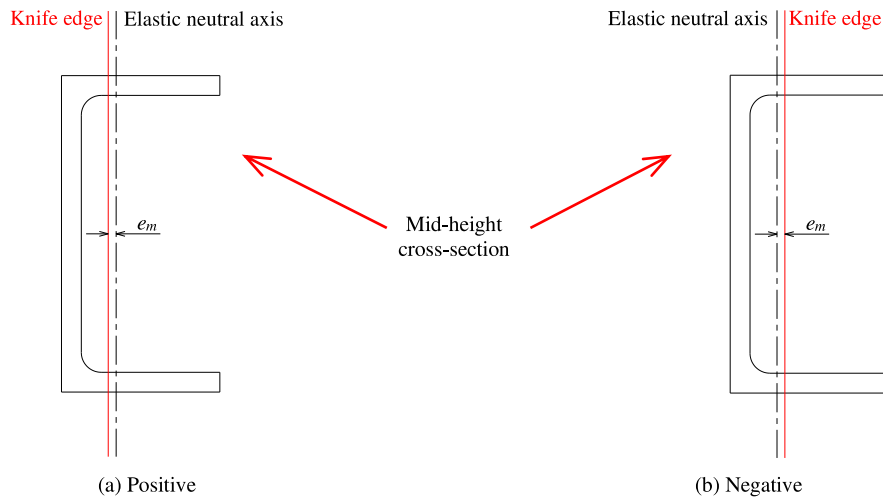


Fig. 13. Pin-ended column test setup.

Fig. 14. Sign convention for overall global geometric imperfection e_m .

and Δ is the mid-height lateral deflection recorded by the LVDT. Note that Eq. (4) was obtained by assuming that the structural behaviour was close to linear elastic and it was recommended that no more than 15% of the expected failure load be used in the determination of e_m [13,32,33]. The overall global geometric imperfection e_m is assumed to be positive if it leads the column specimen to bow towards the flange tips (i.e. with the relative position of the knife edge and mid-height cross-section minor principal axis shown in Fig. 14(a)), while the overall global geometric imperfection e_m is assumed to be negative if it leads the column specimen to bow towards the web (i.e. with the relative position of the knife edge and mid-height cross-section minor principal axis displayed in Fig. 14(b)). It is worth noting that if the absolute value of the overall global geometric imperfection amplitude $|e_m|$ was larger than $L_e/1000$, the position of the column specimen was then carefully re-adjusted until the attainment of $|e_m| < L_e/1000$ [13,33–35]. Table 4 reports the final overall global geometric imperfection amplitude e_m , together with the ratio of its absolute value

to the member effective length $|e_m|/L_e$, for each hot-rolled stainless steel channel section column specimen.

$$e_m = e_0 + \omega_g = \frac{EI(\epsilon_{\max} - \epsilon_{\min})}{Nd_s} - \Delta \quad (4)$$

All the hot-rolled stainless steel channel section column specimens underwent noticeable global deformations upon testing and failed by visible member flexural buckling about the minor principal axis, as shown in Fig. 15(a) and (b). With regard to specimens C2-L1, C2-L4 and C2-L5 with positive overall global geometric imperfection amplitudes, the induced second-order bending moments resulted in tensile stresses at the flange tips and the corresponding failure modes showed minor-axis flexural buckling in the ‘reverse C’ orientation. With regard to the other specimens with negative overall global geometric imperfection amplitudes, the induced second-order bending moments led to compressive stresses at the flange tips and the corresponding failure modes

displayed minor-axis flexural buckling in the 'C' orientation. The load-mid-height lateral deflection curves measured for the C 80 × 40 × 5 and C 100 × 50 × 5 column specimens are shown in Fig. 16(a) and (b), respectively, while the key obtained test results, including the failure load $N_{u, test}$, the mid-height lateral deflection at the failure load Δ_u and the buckling orientation, are reported in Table 4. It is worth noting that the mid-height lateral deflections are taken as negative for channel section column specimens failing by minor-axis flexural buckling in the 'C' orientation, but positive for column specimens failing by minor-axis flexural buckling in the 'reverse C' orientation, which is consistent with the sign convention for initial and overall global geometric imperfections.

3. Numerical modelling

3.1. Overview

Following the testing programme, a numerical modelling programme was conducted by means of the nonlinear finite element (FE) software ABAQUS [36]. FE models were initially developed and validated against the test results and subsequently adopted to perform parametric studies to generate further numerical data on hot-rolled stainless steel channel section columns failing by minor-axis flexural buckling over a wide range of cross-section dimensions and member effective lengths.

3.2. Development of FE models

Each hot-rolled stainless steel channel section column was modelled based on the measured cross-section dimensions and member effective length. The four-node doubly curved shell element allowing finite membrane strains and large rotations, S4R [36], has been widely adopted in previous numerical modelling of metallic channel section members [4,5,7,9,12,13,37] and was thus also employed herein. On the basis of a prior mesh sensitivity study considering mesh sizes varying between $0.2t \times 0.2t$ and $2t \times 2t$, the final element length along the model longitudinal axis and element width along the cross-section centreline were both selected as t ; this size was shown to enable an accurate simulation of membrane residual stresses and also led to a good balance between computational efficiency and accuracy. Regarding the material modelling of hot-rolled stainless steels, the measured (engineering) stress-strain curves were converted into the true stress-plastic strain curves and assigned to the corresponding FE models. The membrane residual stresses were firstly predicted from the predictive model presented in Section 2.4 and then introduced into the FE models by means of the 'Predefined Field' command [36]; Fig. 17 depicts the membrane residual stress pattern and amplitudes incorporated into the FE model for a typical specimen C2-L1.

For ease of application of boundary conditions, each end section of the channel section column FE model was coupled to a concentric reference point; the top reference point was fully restrained except for translation in the longitudinal direction and rotation about the minor principal axis, while the bottom reference point was only allowed for rotation about the minor principal axis, to replicate the same pin-ended boundary conditions as those adopted in the tests. In terms of the incorporation of initial global and local geometric imperfections into each channel section column FE model, the lowest elastic global (minor-axis flexural) and local buckling mode shapes [12,13,32,33,37], obtained from an elastic eigenvalue buckling analysis, were taken as the initial global and local geometric imperfection distribution patterns, respectively; moreover, the lowest elastic global buckling mode shape was oriented such that it was consistent with the corresponding specimen failure orientation. Two local and global imperfection magnitude combinations, including the measured magnitude combination $-\omega_0 + |e_m|$ as well as a generalised magnitude combination $-t/100 + L_e/1000$, were employed to factor the obtained initial geometric imperfection



(a) C 80×40×5 specimens (from left go right: C1-L1, C1-L2, C1-L3, C1-L4 and C1-L5)



(b) C 100×50×5 specimens from left go right: C2-L1, C2-L2, C2-L3, C2-L4 and C2-L5)

Fig. 15. Test failure modes for hot-rolled stainless steel channel section column specimens.

distribution patterns. Upon development of the hot-rolled stainless steel channel section column FE models, the modified Riks method, commonly adopted for solving static numerical problems with material and geometric nonlinearities, was adopted in this study for nonlinear analyses of the developed FE models.

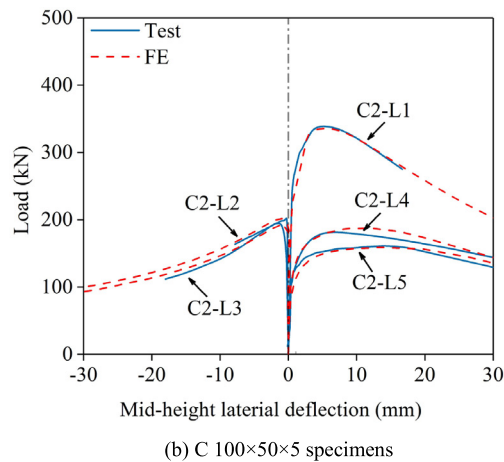
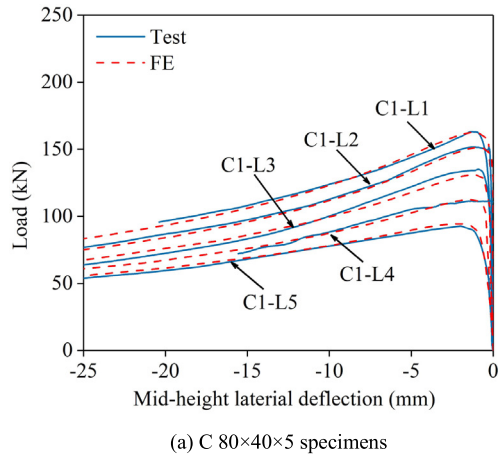
3.3. Validation of FE models

The numerical failure loads, load-mid-height lateral deflection curves and failure modes, as derived from the Riks analyses, were

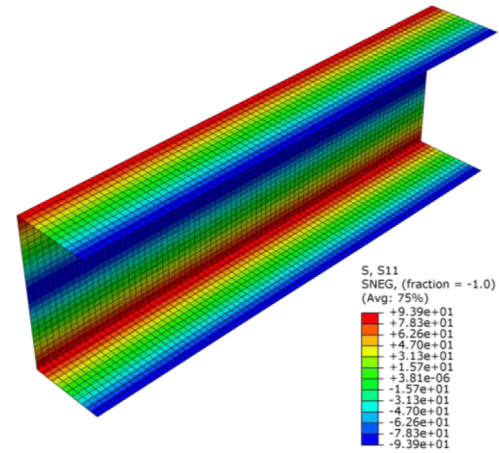
Table 4

Summary of pin-ended hot-rolled stainless steel channel section column test results.

Cross-section	Specimen ID	L_e (mm)	e_m (mm)	$ e_m /L_e$	Failure orientation	$N_{u,test}$ (kN)	Δ_u (mm)
C 80 × 40 × 5	C1-L1	547.4	-0.45	1/1209	C	164.05	-1.34
	C1-L2	748.2	-0.39	1/1916	C	151.51	-1.24
	C1-L3	949.2	-0.57	1/1661	C	135.04	-0.92
	C1-L4	1148.3	-0.52	1/2193	C	111.43	-1.66
	C1-L5	1350.8	-0.48	1/2814	C	92.76	-1.99
C 100 × 50 × 5	C2-L1	650.0	0.42	1/1562	reverse C	338.81	5.21
	C2-L2	848.5	-0.28	1/2995	C	201.83	-0.28
	C2-L3	1046.3	-0.32	1/3244	C	194.77	-1.48
	C2-L4	1270.8	0.25	1/5043	reverse C	181.51	6.83
	C2-L5	1449.5	0.41	1/3501	reverse C	164.14	14.11

**Fig. 16.** Test and FE load-mid-height lateral deflection curves.

compared with those obtained from the tests, allowing the accuracy of the developed FE models to be assessed. Table 5 reports the numerical to test failure load ratios $N_{u,FE}/N_{u,test}$ for ten hot-rolled stainless steel channel section column specimens, with the results revealing that (i) the measured local and global imperfection magnitude combination – $\omega_0 + |e_m|$ yields excellent agreement between the numerical and test failure loads, with the mean numerical to test failure load ratio of 1.00 and the coefficient of variation (COV) of 0.02 and (ii) the generalised local and global imperfection magnitude combination – $t/100 + L_e/1000$ also results in a satisfactory level of test failure load predictions. Fig. 16 presents comparisons between the test and numerical load-mid-height lateral deflection curves for all the hot-rolled stainless steel channel section column specimens, evidencing

**Fig. 17.** Modelled membrane residual stress pattern and amplitudes (in MPa) for specimen C2-L1.**Table 5**

Comparisons of test failure loads with FE failure loads for varying geometric imperfection magnitude combinations.

Cross-section	Specimen ID	$N_{u,FE}/N_{u,test}$	
		$\omega_0 + e_m $	$t/100 + L_e/1000$
C 80 × 40 × 5	C1-L1	0.99	0.98
	C1-L2	1.00	0.93
	C1-L3	0.97	0.89
	C1-L4	1.01	0.87
	C1-L5	1.01	0.85
C 100 × 50 × 5	C2-L1	0.99	0.88
	C2-L2	1.00	0.92
	C2-L3	0.99	0.87
	C2-L4	1.03	0.98
	C2-L5	0.97	0.92
Mean		1.00	0.91
COV		0.02	0.04

that the numerical load-deformation histories well capture their test counterparts. The numerical failure modes were also shown to be in excellent agreement with their test counterparts, as illustrated in Figs. 18 and 19 for typical specimens C1-L3 and C2-L4 with minor-axis flexural buckling in the ‘C’ and ‘reverse C’ orientations. Overall, it can be concluded that the developed FE models can accurately simulate the test responses of the pin-ended hot-rolled stainless steel channel section column specimens and were thus deemed to be validated.

3.4. Parametric studies

Having been validated in Section 3.3, the developed column FE models were adopted to perform parametric studies to generate further numerical data on hot-rolled stainless steel channel section columns

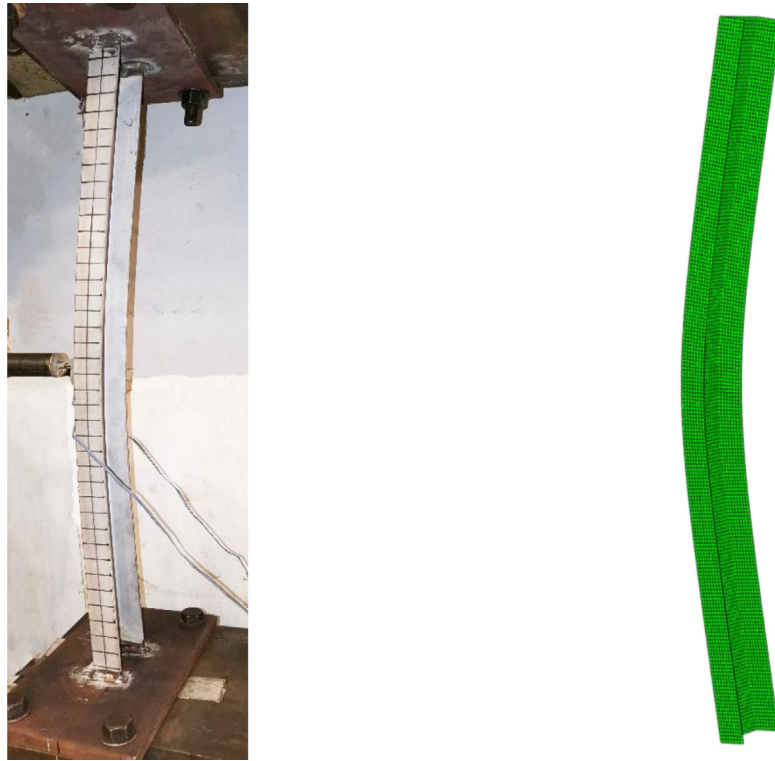


Fig. 18. Test and FE failure modes for specimen C1-L3 (with minor-axis flexural buckling in the 'C' orientation).

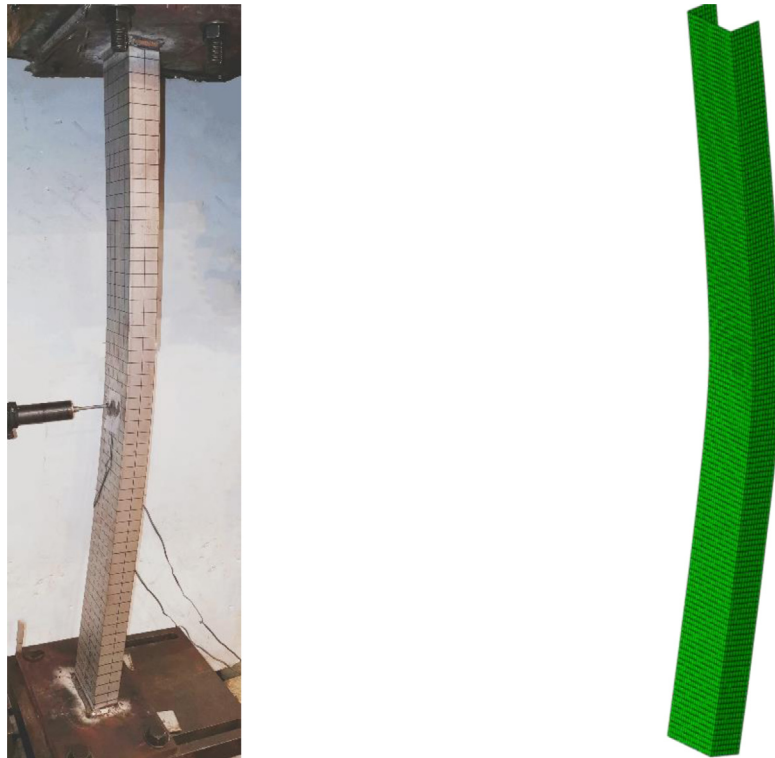


Fig. 19. Test and FE failure modes for specimen C2-L4 (with minor-axis flexural buckling in the 'reverse C' orientation).

over a wide range of cross-section dimensions and member effective lengths. In the parametric studies, the adopted modelling procedures and techniques were the same as those described in Section 3.2, but with the use of the generalised imperfection magnitude combination – $t/100$ and $L_e/1000$ and the material properties of the hot-rolled

stainless steel channel section C 80 × 40 × 5. Table 6 reports the detailed cross-section dimensions and member effective lengths of the modelled channel section columns. Specifically, the outer web widths B_w of all the modelled channel sections were fixed at 180 mm, while three outer flange widths B_f equal to 180 mm, 90 mm and 60 mm were

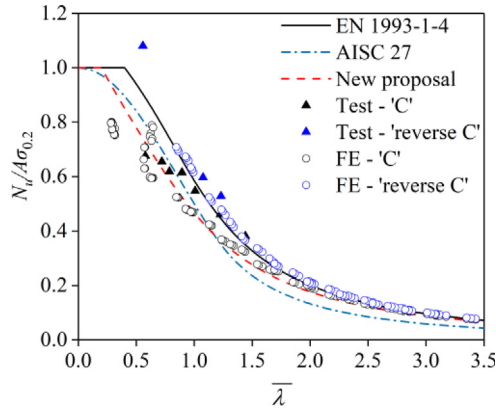


Fig. 20. Comparisons of test and FE failure loads with codified and proposed design buckling curves.

selected, with the wall thickness t varied between 6 mm and 25 mm, leading to three cross-section web-to-flange aspect ratios (1.0, 2.0 and 3.0) and a range of cross-section sizes being considered. It is worth highlighting that (i) the modelled channel sections cover three classes (Class 1–3) of non-slender cross-sections defined in EN 1993-1-4 [16] and are also categorised as non-slender cross-sections prescribed in AISC 27 [17]. For each modelled channel section, ten member effective lengths, ranged from 400 mm to 12000 mm, were selected, to obtain a wide spectrum of member non-dimensional slendernesses $\bar{\lambda}$ from 0.2 to 3.5. A numerical data bank, including 180 data on hot-rolled stainless steel channel section columns failing by minor-axis flexural buckling, has been generated through the parametric studies, with 108 in the 'C' orientation and 72 in the 'reverse C' orientation.

4. Evaluation of existing design methods

4.1. Overview

In this section, the existing design rules for pin-ended hot-rolled stainless steel channel section columns prone to minor-axis flexural buckling, as set out in the European code EN 1993-1-4 [16] and American design guide AISC 27 [17], were discussed, with their accuracy evaluated against the obtained test and numerical data. Table 7 reports the quantitative evaluation results, including the mean test and numerical to predicted failure load ratios and the corresponding COVs, while the graphical evaluation results are shown in Figs. 20–23. Based on the quantitative and graphical evaluation results, the shortcomings of the codified design rules were discussed and highlighted. Finally, a new improved design approach was proposed for hot-rolled stainless steel channel section columns prone to minor-axis flexural buckling.

4.2. EN 1993-1-4 (EC3)

The European code EN 1993-1-4 [16] provides supplementary design rules for stainless steel structures in addition to the general design rules for mild steel structures set out in EN 1993-1-1 [38]. With regard to the design of column members failing by global buckling (e.g., flexural, torsional and flexural–torsional buckling), the concept of buckling curves, as developed based on the Perry–Robertson buckling formula, is adopted. Four buckling curves are defined in EN 1993-1-4 [16] and reflect the different effects of initial geometric imperfections and residual stresses on the column buckling resistances. The unfactored EC3 design resistance of a column failing by global buckling $N_{u,EC3}$ is calculated from Eq. (5), where χ is the reduction factor and determined from the EC3 design buckling curves, as given by Eq. (6), where ϕ is a buckling parameter, as calculated from Eq. (7), where α is the imperfection factor and $\bar{\lambda}_0$ is the limiting slenderness; note that the

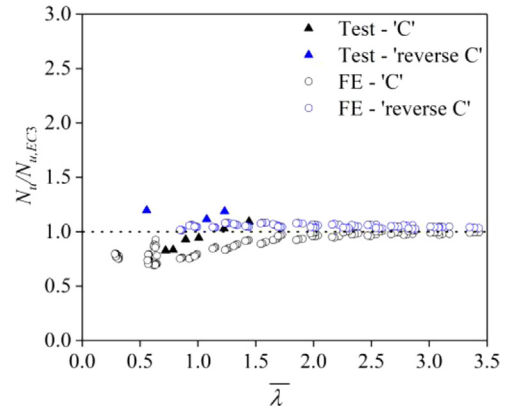


Fig. 21. Comparisons of test and FE failure loads with EC3 flexural buckling resistance predictions.

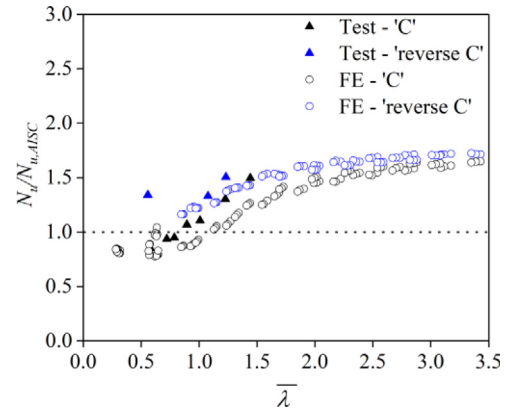


Fig. 22. Comparisons of test and FE failure loads with AISC flexural buckling resistance predictions.

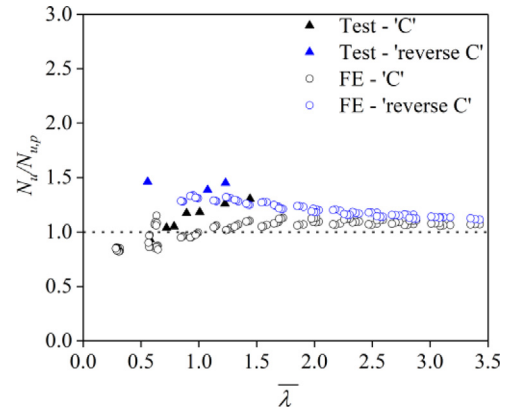


Fig. 23. Comparisons of test and FE failure loads with flexural buckling resistance predictions from new design approach.

values of α and $\bar{\lambda}_0$ are equal to 0.49 and 0.40, respectively, for stainless steel channel section columns failing by flexural buckling about the minor principal axis.

$$N_{u,EC3} = \chi A \sigma_{0.2} \quad (5)$$

$$\chi = \frac{1}{\phi + \sqrt{\phi^2 - \bar{\lambda}^2}} \leq 1 \quad (6)$$

$$\phi = 0.5[1 + \alpha(\bar{\lambda} - \bar{\lambda}_0) + \bar{\lambda}^2] \quad (7)$$

Table 6

Cross-section dimensions and member effective lengths selected for parametric studies.

B_w (mm)	B_f (mm)	t (mm)	r (mm)	L_e (mm)
180	60	9	9	400, 800, 1200, 1600, 2000, 2400, 2800, 3200, 3600, 4000
180	60	7	7	400, 800, 1200, 1600, 2000, 2400, 2800, 3200, 3600, 4000
180	60	6	6	400, 800, 1200, 1600, 2000, 2400, 2800, 3200, 3600, 4000
180	90	13	13	600, 1300, 2000, 2700, 3400, 4100, 4800, 5500, 6200, 6900
180	90	11	11	600, 1300, 2000, 2700, 3400, 4100, 4800, 5500, 6200, 6900
180	90	9	9	600, 1300, 2000, 2700, 3400, 4100, 4800, 5500, 6200, 6900
180	180	25	25	1200, 2400, 3600, 4800, 6000, 7200, 8400, 9600, 10800, 12000
180	180	22	22	1200, 2400, 3600, 4800, 6000, 7200, 8400, 9600, 10800, 12000
180	180	18	18	1200, 2400, 3600, 4800, 6000, 7200, 8400, 9600, 10800, 12000

Table 7

Comparisons of test and FE failure loads with predicted flexural buckling resistances.

Failure orientation	No. of data		$N_u/N_{u,EC3}$		$N_u/N_{u,AISC}$		$N_u/N_{u,p}$	
	Test	FE	Mean	COV	Mean	COV	Mean	COV
C	7	108	0.88	0.11	1.19	0.32	1.02	0.12
reverse C	3	72	1.06	0.03	1.51	0.20	1.22	0.08
Total	10	180	0.95	0.12	1.33	0.31	1.10	0.14

The EC3 flexural buckling resistance predictions $N_{u,EC3}$ were evaluated against the test and numerical failure loads N_u . The graphical evaluation results are displayed in Fig. 20, where the test and numerical failure loads N_u , categorised by failure orientation, are normalised by the cross-section yield loads $A\sigma_{0.2}$ and plotted against the member non-dimensional slendernesses $\bar{\lambda}$, along with the EC3 design buckling curve, and in Fig. 21, where the test and numerical to EC3 predicted failure load ratios $N_u/N_{u,EC3}$, categorised by failure orientation, are plotted against the member non-dimensional slendernesses $\bar{\lambda}$. It can be observed that (i) the test and numerical data points for hot-rolled stainless steel channel section columns failing in the 'C' orientation lie below those for hot-rolled stainless steel channel section columns failing in the 'reverse C' orientation, which is attributed to the fact that the second-order bending moment associated with 'C'-orientation flexural buckling leads to compressive stresses at the flange tips and hence would accelerate the global instability; (ii) the EC3 design buckling curve generally yields safe-sided, accurate and consistent flexural buckling resistance predictions for hot-rolled stainless steel channel section columns failing in the 'reverse C' orientation; (iii) the EC3 design buckling curve, however, leads to a rather large number of unsafe-sided flexural buckling resistance predictions for hot-rolled stainless steel channel section columns with failure in the 'C' orientation, especially those falling within the intermediate and low member non-dimensional slenderness range. The above analyses and discussions are also supported by the quantitative evaluation results reported in Table 7, where the mean ratios of $N_u/N_{u,EC3}$ are equal to 0.88 and 1.06 for hot-rolled stainless steel channel section columns failing by minor-axis flexural buckling in the 'C' and 'reverse C' orientations, with the corresponding COVs of 0.11 and 0.03.

4.3. AISC 27 (AISC)

The American design guide AISC 27 [17] uses the critical stress approach to determine the strengths of hot-rolled stainless steel structural columns. For non-slender element compression members failing by flexural buckling, the design member strength $N_{u,AISC}$ is determined as the product of the critical stress F_{cr} and the gross cross-section area A , as shown by Eq. (8), where the critical stress F_{cr} is dependent on the member non-dimensional slenderness $\bar{\lambda}$ and determined from a buckling curve defined by Eq. (9).

$$N_{u,AISC} = AF_{cr} \quad (8)$$

$$F_{cr} = \begin{cases} 0.50\bar{\lambda}^2 \sigma_{0.2} & \text{for } \bar{\lambda} \leq 1.2 \\ 0.531 \frac{\sigma_{0.2}}{\bar{\lambda}^2} & \text{for } \bar{\lambda} > 1.2 \end{cases} \quad (9)$$

The design flexural buckling strengths $N_{u,AISC}$ of hot-rolled stainless steel channel section columns, as predicted from AISC 27 [17], were assessed through comparisons against the test and numerical failure loads N_u . The quantitative assessment results are reported in Table 7, where the mean ratios of $N_u/N_{u,AISC}$ are equal to 1.19 and 1.51, for hot-rolled stainless steel channel section columns with failure in the 'C' and 'reverse C' orientations, respectively, with the corresponding COVs of 0.32 and 0.20, indicating that AISC 27 [17] yields unduly conservative and scattered flexural buckling strength predictions for hot-rolled stainless steel channel section columns. The conservatism and scatter of the AISC flexural buckling strength predictions are also evident in Fig. 22, where the ratios of $N_u/N_{u,AISC}$ are plotted against the member non-dimensional slendernesses $\bar{\lambda}$. In comparison with the EC3 design buckling curve, the AISC design buckling curve results in more conservative and scattered flexural buckling strength predictions for hot-rolled stainless steel channel section columns over the whole considered range of member non-dimensional slendernesses up to 3.5, but with a significantly reduced number of unsafe strength predictions.

4.4. New design approach

In view of the fact that (i) the EC3 design buckling curve yields a rather large number of unsafe-sided flexural buckling resistance predictions for hot-rolled stainless steel channel section columns with member non-dimensional slendernesses up to around 1.7 and (ii) the AISC critical stress approach greatly underestimates the flexural buckling resistances, there is a necessity to develop a new, more accurate, design method. In this section, a new design method was developed based on the original EC3 design buckling curve, but with the use of the revised limiting slenderness $\bar{\lambda}_0$ equal to 0.2 and revised imperfection factor α equal to 0.76. The predicted flexural buckling resistances $N_{u,p}$ were now calculated from the revised EC3 design buckling curve and compared with the test and numerical failure loads N_u . The graphical and quantitative comparison results are presented in Figs. 20 and 23 and Table 7, respectively. In comparison with the original EC3 design buckling curve, the revised EC3 design buckling curve leads to a greatly reduced number of unsafe flexural buckling resistance predictions for hot-rolled stainless steel channel section columns; moreover, the revised EC3 design buckling curve was also shown to offer a much higher degree of design accuracy and consistency than the AISC critical stress approach.

The reliability of the revised EC3 design buckling curve was assessed through statistical analyses, according to the requirements of EN 1990 [39]. In the reliability analyses, the material over-strength ratio for austenitic stainless steel was taken as 1.30 with the COV of

Table 8

Reliability analysis results calculated according to EN 1990 [39].

Design method	No. of test and FE data	$k_{d,n}$	b	V_δ	V_r	γ_{M1}
Revised EN 1993-1-4	190	3.20	1.09	0.09	0.12	1.07
Original EN 1993-1-4	190	3.20	0.91	0.09	0.12	1.27

0.06 [40], while the COV of the geometric properties of stainless steel cross-sections was equal to 0.05 [40]. The key statistical parameters, calculated based on a total of 190 test and numerical data, are reported in Table 8, where $k_{d,n}$ is the design (ultimate limit state) fractile factor, b is the mean ratio of test and numerical resistances to design model resistances, V_δ is the COV of the test and numerical resistances relative to the resistance model, V_r is combined COV incorporating both model and basic variable uncertainties, and γ_{M1} is the partial safety factor for stainless steel member resistance. As reported in Table 8, the resulting (calculated) partial safety factor for the revised EC3 design buckling curve is equal to 1.07, which is less than the currently adopted value of 1.10 in EN 1993-1-4 [16]. This demonstrates that the revised EC3 design buckling curve for pin-ended hot-rolled stainless steel channel section columns failing by minor-axis flexural buckling satisfies the reliability requirements of EN 1990 [39].

5. Conclusions

The minor-axis flexural buckling behaviour and resistances of pin-ended hot-rolled stainless steel channel section columns have been investigated through a thorough testing and numerical modelling programme. The testing programme included material tensile coupon tests, initial global and local geometric imperfection measurements, membrane residual stress measurements and compression tests on ten pin-ended hot-rolled stainless steel channel section column specimens about the minor principal axis. The test setup and procedures were reported in detail. The key obtained test results, including the failure loads, load–mid-height lateral deflection curves and failure modes with minor-axis flexural buckling in the ‘C’ and ‘reverse C’ orientations, were presented and discussed. The test results were used in a parallel numerical modelling programme for validating the developed FE models, which were then adopted to perform parametric studies to generate further numerical data over a wide range of cross-section dimensions and member effective lengths. The test and numerical data were used to assess the accuracy of the codified design buckling curves, as set out in EN 1993-1-4 [16] and AISI 27 [17], for pin-ended hot-rolled stainless steel channel section columns prone to flexural buckling about the minor principal axis and develop new improved design buckling curve. The following conclusions can be made:

- The EC3 design buckling curve was found to lead to many unsafe flexural buckling resistance predictions for hot-rolled stainless steel channel section columns with failure in the ‘C’ orientation.
- The AISI design buckling curve results in rather conservative and scattered resistance predictions.
- A revised EC3 design buckling curve was proposed with the imperfection factor $\alpha = 0.76$ and limiting slenderness $\bar{\lambda}_0 = 0.2$, and shown to offer accurate, consistent, safe-sided and reliable flexural buckling resistance predictions for pin-ended hot-rolled stainless steel channel section columns.

CRediT authorship contribution statement

Shuai Li: Conceptualization, Formal analysis, Investigation, Methodology, Visualization, Writing - original draft. **Lulu Zhang:** Investigation, Methodology, Writing - original draft. **Yating Liang:** Conceptualization, Formal analysis, Investigation, Methodology, Supervision, Writing - review & editing. **Ou Zhao:** Conceptualization, Formal analysis, Funding acquisition, Investigation, Methodology, Resources, Supervision, Writing - review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors thank Stainless Structurals Asia for sponsoring hot-rolled stainless steel channel section columns. The assistances from Mr Lee Guangjie, Mr Chelladurai Subasanran and Mr Lim Yong Cheng during the experiments are also acknowledged.

References

- [1] N.R. Baddoo, Stainless steel in construction: A review of research, applications, challenges and opportunities, *J. Construct. Steel Res.* 64 (11) (2008) 1199–1206.
- [2] L. Gardner, Stability and design of stainless steel structures – review and outlook, *Thin-Walled Struct.* 141 (2019) 208–216.
- [3] H. Kuwamura, Local buckling of thin-walled stainless steel members, *Steel Struct.* 3 (3) (2003) 191–201.
- [4] J. Dobrić, D. Budevac, Z. Marković, N. Gluhović, Behaviour of stainless steel press-braked channel sections under compression, *J. Construct. Steel Res.* 139 (2017) 236–253.
- [5] L. Zhang, K.H. Tan, O. Zhao, Local stability of press-braked stainless steel angle and channel sections: Testing, numerical modelling and design analysis, *Eng. Struct.* 203 (2020) 109869.
- [6] M. Theofanous, A. Liew, L. Gardner, Experimental study of stainless steel angles and channels in bending, *Structures* 4 (2015) 80–90.
- [7] Y. Liang, O. Zhao, Y. Long, L. Gardner, Stainless steel channel sections under combined compression and minor axis bending–Part 1: Experimental study and numerical modelling, *J. Construct. Steel Res.* 152 (2019) 154–161.
- [8] Y. Liang, O. Zhao, Y. Long, L. Gardner, Stainless steel channel sections under combined compression and minor axis bending–Part 2: Parametric studies and design, *J. Construct. Steel Res.* 152 (2019) 162–172.
- [9] Y. Liang, O. Zhao, Y.L. Long, et al., Experimental and numerical studies of laser-welded stainless steel channel sections under combined compression and major axis bending moment, *Thin-Walled Struct.* 157 (2020) 107035.
- [10] J. Becque, K.J.R. Rasmussen, Experimental investigation of local–overall interaction buckling of stainless steel lipped channel columns, *J. Construct. Steel Res.* 65 (8–9) (2009) 1677–1684.
- [11] B. Rossi, J.P. Jaspart, K.J.R. Rasmussen, Combined distortional and overall flexural–torsional buckling of cold-formed stainless steel sections: experimental investigations, *J. Struct. Eng.* 136 (4) (2009) 354–360.
- [12] J. Dobrić, J. Ivanović, B. Rossi, Behaviour of stainless steel plain channel section columns, *Thin-Walled Struct.* 148 (2020) 106600.
- [13] L. Zhang, K.H. Tan, O. Zhao, Press-braked stainless steel channel section columns failing by flexural buckling: Testing, numerical simulation and design, *Thin-Walled Struct.* (2020).
- [14] Montanstahl special profiles in steel. <https://www.montanstahl.com>.
- [15] Stainless Structurals Asia. <https://www.montanstahl.com/contact/singapore>.
- [16] EN 1993-1-4:2006+A1:2015. Eurocode 3: Design of Steel Structures – Part 1.4: General Rules – Supplementary Rules for Stainless Steels, Including Amendment A1 (2015), European Committee for Standardization (CEN), Brussels, 2015.
- [17] AISI Design Guide 27, Structural Stainless Steel, American Institute of Steel Construction (AISC), 2013.
- [18] ISO EN. 6892-1. Metallic Materials–Tensile Testing–Part 1: Method of Test At Room Temperature, International Organization for Standardization, 2009.
- [19] W. Ramberg, W.R. Osgood, Description of stress–strain curves by three parameters, 1943.
- [20] H.N. Hill, Determination of Stress–Strain Relations from Offset Yield Strength Values, Aluminum Co Of America Pittsburgh Pa, 1944.
- [21] E. Mirambell, E. Real, On the calculation of deflections in structural stainless steel beams: an experimental and numerical investigation, *J. Construct. Steel Res.* 54 (1) (2000) 109–133.
- [22] K.J.R. Rasmussen, Full-range stress–strain curves for stainless steel alloys, *J. Constr. Steel Res.* 59 (1) (2003) 47–61.
- [23] I. Arrayago, E. Real, L. Gardner, Description of stress–strain curves for stainless steel alloys, *Mater. Des.* 87 (2015) 540–552.
- [24] ES. dos Santos, EM. Batista, D. Camotim, Experimental investigation concerning lipped channel columns undergoing local–distortional–global buckling mode interaction, *Thin-Walled Struct.* 54 (2012) 19–34.
- [25] B. Young, P.B. Dinis, D. Camotim, CFS Lipped channel columns affected by L–D–G interaction, Part I: Experimental investigation, *Comput. Struct.* 207 (2018) 219–232.
- [26] R. Rinchen, K.J.R. Rasmussen, Experiments on long-span cold-formed steel single C-section portal frames, *J. Struct. Eng.* 146 (1) (2020) 04019187.

- [27] J. Ye, I. Hajirasouliha, J. Becque, Experimental investigation of local-flexural interactive buckling of cold-formed steel channel columns, *Thin-Walled Struct.* 125 (2018) 245–258.
- [28] B. Schafer, T. Peköz, Computational modeling of cold-formed steel: characterizing geometric imperfections and residual stresses, *J. Constr. Steel Res.* 47 (1998) 193–210.
- [29] RD. Ziemian, *Guide to Stability Design Criteria for Metal Structures*, John Wiley & Sons, 2010.
- [30] N. Tebedge, G. Alpsten, L. Tall, Residual-stress measurement by the sectioning method, *Exp. Mech.* 13 (2) (1973) 88–96.
- [31] Y. Huang, B. Young, Experimental investigation of cold-formed lean duplex stainless steel beam-columns, *Thin-Walled Struct.* 76 (2014) 105–117.
- [32] C. Buchanan, E. Real, L. Gardner, Testing, simulation and design of cold-formed stainless steel CHS columns, *Thin-Walled Struct.* 130 (2018) 297–312.
- [33] F. Wang, Y. Liang, O. Zhao, Experimental and numerical studies of pin-ended press-braked S960 ultra-high strength steel channel section columns, *Eng. Struct.* 215 (2020) 110629.
- [34] KJR. Rasmussen, GJ. Hancock, Tests of high strength steel columns, *J. Constr. Steel Res.* 34 (1995) 27–52.
- [35] L. Gardner, Y. Bu, M. Theofanous, Laser-welded stainless steel I-sections: Residual stress measurements and column buckling tests, *Eng. Struct.* 127 (2016) 536–548.
- [36] ABAQUS, *ABAQUS/Standard User's Manual*, Version 6.17, Dassault Systemes Simulia Corp, USA, 2017.
- [37] J. Becque, K.J.R. Rasmussen, A numerical investigation of local-overall interaction buckling of stainless steel lipped channel columns, *J. Construct. Steel Res.* 65 (8–9) (2009) 1685–1693.
- [38] EN 1993-1-1. *Eurocode 3: Design of Steel Structures – Part 1–1: General Rules and Rules for Buildings*, CEN, Brussels (Belgium), 2005.
- [39] EN. *Eurocode – Basis of Structural Design*, European Committee for Standardization (CEN), Brussels, 1990, p. 2002.
- [40] S. Afshan, P. Francis, N.R. Baddoo, et al., Reliability analysis of structural stainless steel design provisions, *J. Construct. Steel Res.* 114 (2015) 293–304.